

module3_lab

September 17, 2026

```
[1]: import pandas as pd
df = pd.read_csv("messy_sales.csv")
```

```
[2]: df.head(10)
```

```
[2]:
```

	order_id	date	product	price	qty	zip
0	1254	04/06/2026	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	05/06/2026	mug	5.50	1	10001
3	1066	05/30/2026	notebook	NaN	1	98101
4	1114	04/06/2026	webcam	45.59	4	90405
5	1280	04/22/2026	pen set	NaN	4	30303
6	1204	04/18/2026	charger	8.41	2	2134
7	1249	2026-05-15	keyboard	75.04	4	90405
8	1037	2026-05-19	mug	9.31	2	60614
9	1177	2026-05-13	mug	10.11	3	90210

```
[3]: df.shape
```

```
[3]: (300, 6)
```

```
[4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    300 non-null   int64
1   date        300 non-null   object
2   product     300 non-null   object
3   price       288 non-null   float64
4   qty         300 non-null   int64
5   zip         300 non-null   int64
dtypes: float64(1), int64(3), object(2)
memory usage: 14.2+ KB
```

```
[5]: df.describe()
```

```
[5]:
```

	order_id	price	qty	zip
count	300.000000	288.000000	300.000000	300.000000
mean	1144.816667	56.068438	2.913333	45385.276667
std	84.515319	72.307182	1.560107	39345.608641
min	1000.000000	2.930000	-5.000000	2116.000000
25%	1071.750000	10.945000	2.000000	2134.000000
50%	1144.500000	37.530000	3.000000	30303.000000
75%	1217.250000	61.417500	4.000000	90210.000000
max	1291.000000	479.290000	5.000000	98101.000000

0.0.1 Intial Obserations

Three things that I see wrong with this data set is: 1. The price count is 288 but there are 300 rows, so there may be missing values. 2. The minimum for quantity is -5 which seems suspicious for a sales order. 3. The date column has inconsistent formats, some follow MM/DD/YYYY and others YYYY-MM-DD.

```
[6]: df.isna().sum()
```

```
[6]: order_id    0
      date       0
      product    0
      price     12
      qty        0
      zip        0
      dtype: int64
```

```
[7]: fill_value = df["price"].median()
      df["price"] = df["price"].fillna(fill_value)
```

```
[8]: print (fill_value)
```

```
37.53
```

```
[9]: df.isna().sum()
```

```
[9]: order_id    0
      date       0
      product    0
      price     0
      qty        0
      zip        0
      dtype: int64
```

```
[10]: print(df.duplicated().sum())
```

```
8
```

```
[11]: df.shape
```

```
[11]: (300, 6)
```

```
[12]: df = df.drop_duplicates()
```

```
[13]: df.shape
```

```
[13]: (292, 6)
```

```
[14]: df["zip"] = df["zip"].astype(str).str.zfill(5)
```

```
[15]: df.head(10)
```

```
[15]:
```

	order_id	date	product	price	qty	zip
0	1254	04/06/2026	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	05/06/2026	mug	5.50	1	10001
3	1066	05/30/2026	notebook	37.53	1	98101
4	1114	04/06/2026	webcam	45.59	4	90405
5	1280	04/22/2026	pen set	37.53	4	30303
6	1204	04/18/2026	charger	8.41	2	02134
7	1249	2026-05-15	keyboard	75.04	4	90405
8	1037	2026-05-19	mug	9.31	2	60614
9	1177	2026-05-13	mug	10.11	3	90210

```
[16]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 292 entries, 0 to 299
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0  order_id    292 non-null    int64
 1  date        292 non-null    object
 2  product     292 non-null    object
 3  price       292 non-null    float64
 4  qty         292 non-null    int64
 5  zip         292 non-null    object
dtypes: float64(1), int64(2), object(3)
memory usage: 16.0+ KB
```

```
[17]: df["date"] = pd.to_datetime(df["date"], format="mixed")
```

```
[18]: df.dtypes
```

```
[18]: order_id      int64
      date       datetime64[ns]
      product    object
      price      float64
```

```
qty                int64
zip                object
dtype: object
```

```
[ ]:
```

0.1 Checkpoint

```
[19]: df.shape
```

```
[19]: (292, 6)
```

```
[20]: df.head(10)
```

```
[20]:
```

	order_id	date	product	price	qty	zip
0	1254	2026-04-06	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	2026-05-06	mug	5.50	1	10001
3	1066	2026-05-30	notebook	37.53	1	98101
4	1114	2026-04-06	webcam	45.59	4	90405
5	1280	2026-04-22	pen set	37.53	4	30303
6	1204	2026-04-18	charger	8.41	2	02134
7	1249	2026-05-15	keyboard	75.04	4	90405
8	1037	2026-05-19	mug	9.31	2	60614
9	1177	2026-05-13	mug	10.11	3	90210

```
[21]: df.dtypes
```

```
[21]: order_id          int64
date              datetime64[ns]
product           object
price             float64
qty              int64
zip              object
dtype: object
```

```
[22]: df[df["qty"] < 0]
```

```
[22]:
```

	order_id	date	product	price	qty	zip
202	1140	2026-04-18	webcam	38.69	-5	98101
262	1233	2026-05-09	desk lamp	22.64	-4	10001
297	1025	2026-04-27	keyboard	47.30	-2	02116

```
[23]: df.to_csv("sales_clean.csv", index=False)
```

0.2 Cleaning Log

- There was a total of 300 rows loaded

- Filled 12 missing price values with median 37.53.
- The maximum price is 479.29, the mean is 56.07 and median is 37.53. I chose the median because the high price can skew the data and mean.
- Removed 8 duplicate rows leaving 292.
- Converted zip codes from integers to string and restored leading zeroes by adapting to 5-character zips.
- Converted the date column from object to real datetimes and standardized the mixed formats.
- Decided to treat the 3 negative quantities (-5,-4,-2) as possible returns instead of typos. I decided to keep these the rows instead of correcting or deleting, since return of items is possible.
- A wrong choice at any of these steps could mislead a business decision because it could give a false idea of revenue or even the amount of a product that is sold. For example, if I decided to keep the duplicated rows that would increase the total revenue and product quantity.

Git Hub Link: <https://github.com/jazzminowens/cs82a-portfolio>

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